



Deckblatt Auftragsdokumentation

Coverage Documentation

Coverture Documentation

Gegenstand OBJECT OBJET	:	Federungszyylinder CYLINDER SPRING ASSY. CYLINDRE DE SUSPENSION
Abmessung MEASURE MÉTRAGE	:	D 380/160-280 Hub / stroke /course
Loesche Bestell-Nr. LOESCHE ORDER-NO. LOESCHE COMMANDE-NO.	:	4501.7985
Kennwort REFERENCE	:	Narayangonj
Hersteller Auftrags-Nr. MANUFACTURE ORDER-NO. REF. NO. DU MANUFACTEUR	:	107215
Erstellungsdatum MANUFACTURING DATE DATE DE MANUFACTURATION	:	2004

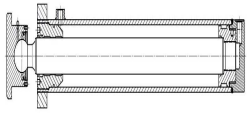
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ASSEMBLY, START-UP AND MAINTENANCE
INSTRUCTION DE MONTAGE

Ersatzteilliste							
Spare Parts List							
Bezeichnung :		Federungszyylinder / Cylinder spring assy.					
Type							
Abmessung :		D 380/160 - 280 Hub / stroke			Kennwort:		Narayangonj
Measure					Key Word		
RUHFUS-Zeichnungs Nr.:		50-1662-1-0					
RUHFUS-Drawing-No.							
Pos.	St.	Bezeichnung	Zeichnungs-Nr.	Abmessung	Qualität	DIN / EN	Bemerkungen
Pos.	Piec.	Name	Drawing-No.	Measure	Quality	DIN / EN	Remarks
1	1	Zylinderrohr kpl. Cyl. Tube assy.	1587.01.00.00.00.1/0				
2	1	Kolbenstange kpl. Piston od cpl.	1662.02.00.00.00.2/0				
9	1	Zylinderkopf kpl. Cylinder head cpl.	1587.09.00.00.00.3/0				
14	1	Zwischenboden Connection bottom	1587.14.00.00.00.2/0				
15	1	Bodenaug Bottom eye	1587.15.00.00.00.1/0				
18	2	Deckel Cover	1587.18.00.00.00.2/0				
25	1	Flachschmierkopf Mini measuring device		R 1/4		3404	
26	2	Meßanschluß Mini measuring device	EMA3-M12x1,5-ED				
27	1	Gereade E.-verschrbg. Safty piece	GE 25-PSR-ED				
28	1	Entlüftungsrohr Tube clamps	1587.28.00.00.00.X/0	D 25x2,5x640			
29	2	Ringschraube Eye srew		M24		580	
30	24	Sechskantschraube Screw		M20x100	10.9	931	
31	48	HV-Scheibe HV-Washer		D 21	St	6916	
32	8	Zylinderschraube Screw		M12x25	8.8	912	
33	8	Federring Spring washer		A12	Fed.St.	7980	
34	24	Sechskantschraube Screw		M20x160	10.9	931	
35	3	Verschlußschraube Lock screw	VSTI-R1/2-ED	G 1/2			
36	1	Hochdruckschlauch Hose	HFM SKE2000				
37	4	Zylinderschraube Screw		M16x65	10.9	912	
38	1	Einstb. Winkel-Versbg. Adjustable male stud elbow	EVW 12-PLR				
39	1	Gereade E.-verschrbg. Male Straight Union	GE 12-PLR-ED				
40	1	Dichtsatz kpl. Sealing set cpl.	1559.EDS				
40.1	1	Abstreifer Scraper ring		D 160/175x10,1	PTFE-Bz/NBR		

Ersatzteilliste							
Spare Parts List							
Bezeichnung :		Federungszyylinder / Cylinder spring assy.					
Type							
Abmessung :		D 380/160 - 280 Hub / stroke			Kennwort:	Anshan	
Measure					Key Word		
RUHFUS-Zeichnungs Nr.:		50-1662-1-0					
RUHFUS-Drawing-No.							
Pos.	St.	Bezeichnung	Zeichnungs-Nr.	Abmessung	Qualität	DIN / EN	Bemerkungen
Pos.	Piec.	Name	Drawing-No.	Measure	Quality	DIN / EN	Remarks
40.2	1	Stangendichtung		D 160/180x16	PU		
		Rod seal					
40.3	1	Dachmanschette		D 160/180x40	NBR / Gewebe		
		Chevron sael					
40.4	1	Stützring	1559.40.04.00.00.4/0	D 160/180x6	PTFE-Bz/NBR		
		Back Ring					
40.5	3	Stangenführungsring		D 160/165x25	Hartgewebe		
		Piston rod wear ring					
40.6	1	Kolbendichtung		D 380/352x9,5	mit Eckversteifung		
		Piston seal					
40.7	3	Kolbenführungsring		D 380/375x25	Hartgewebe		
		Piston wear ring					
40.8	2	O-Ring		D 365x7	NBR 70		
		O-Seal					
40.9	2	Stützring		D 383/371,4x1,7	POM		
		Backring					
51	2	Blockkugelhahn	BKHL 10 L	NW10, pmax.=500 bar			
		Ball valve					
52	1	Ger. Einschraubstutzen	EVGE12 LR-ED				
		Male stud coupling					
53	2	Überwurfmutter	M12-L	M18x1,5			
		Nut					
54	2	VKA-Verschlußkegel	VKA12				
		Blanking plug					
55	4	Sechskantschraube		M5x10	8.8	933	
		Screw					
57	1	Gerade Zwischenstutzen	GZ 12-L				
		Male Couplings					
58	1	Verschlußschraube	VSTI-M12x1,5-ED	M12x1,5			
		Lock screw					
60	24	Sechskantschraube		M12x25	8.8	933	
		Screw					
61	4	Sechskantschraube		M8x16	8.8	933	
		Screw					
63	1	Pendelrollenlager	Beistellung Fa. LOESCHE				
		Saffty holder					
65	4	Abdeckblech	1428.65.00.00.00.3/0				
		Cover					
70	2	Deckel	1587.70.00.00.00.4/0				

		Cover cpl.					
.							

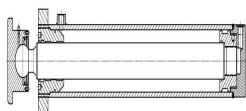


Assembly,
Start-up and Maintenance
of Hydraulic Cylinders

Data Sheet No.
HY-3910-4
Issued: 11/2002

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Assembly, Start-up and Maintenance of Hydraulic Cylinders

Data Sheet No.
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1. General Description

Before starting up the hydraulic cylinder consult the following standards:

- | | |
|----------------------|----------------------|
| - German standards | DIN 24346, para. 5.2 |
| - ISO standard | ISO 4413, para. 6.2 |
| - European standards | EN 982, para. 5.3.2 |

2. Mounting and Assembly

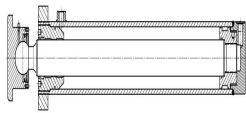
- Cleanliness:

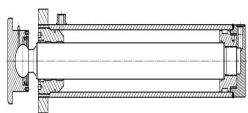
- When mounting the cylinder and in the vicinity
- of pipes and containers all traces of soiling, scales, sand, chips, etc. are to be cleaned before starting assembly
- hot-bent or welded pipes are to be finally pickled, cleaned and oiled.
- When cleaning use only non-linting cloths or special paper
- Do not use sealants such as hemp, putty or sealing tape

3. Start-up

3.1 Pressure Liquid

- Observe the set pressure and temperature ranges.
- Check whether the pressure liquid in the system matches the pressure liquid recommended for the hydraulic cylinder.
- The maximum temperature recommended by the oil manufacturer must not be exceeded.
In order to assure the system's constant operating performance, it is recommended to keep a constant pressure liquid temperature ($\pm 5^{\circ}\text{C}$).

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	• As a rule mineral oils (HL, HLP) as per DIN 51524 can be used Other pressure liquids only at request. 3.2 Filtering • A reliable form of filtering can increase the hydraulic cylinder's service life. Follow the recommendations for the maximum permissible fouling factor for the pressure liquid as per NAS 1638 / ISO DIS 4406. • Filters with electric soiling indicators are to be preferred. • When changing the filter make sure to provide for the cleanliness. 3.3 Bleeding • At idle run pressure loosen the bleeder screw or the bottom- or rod-side screw. • Let the air escape • Tighten the bleeder screw or the other relevant screws only when the outflowing oil is bubble-free. 4. Maintenance • Generally speaking, the hydraulic cylinder does not require any maintenance; make sure to lubricate the various bearing such as the swivel pins as well as the spindle carriers and ball-and-socket joints. • Check for the leak-proof closure at brief intervals • The motion seals are parts subject to wearing. Should the internal or external leakage become too great, we recommend that you change the seals or send the hydraulic cylinder to our factory so that the guides and the other parts can be checked during the changing of the seals.	



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5. Usage limitations for sealing and guide elements

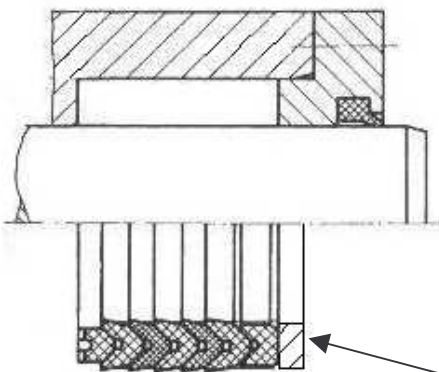
- Temperature range -20°C to $+80^{\circ}\text{C}$
- Max. sliding speeds
 - Chevron packings; grooved rings : 0.5 m/sec
 - Glydring seals : up to 15 m/sec
 - Scraper : 4 – 15 m/sec.
(manufacturer-specific)
 - Guide elements : min. 1 m/sec or higher
(manufacturer-specific)
- You can find the various pressure ratings and proof pressures in the assembly drawings.

6. Assembly of seals / scrapers / guide elements

6.1 Rod seals

6.1.1 Chevron packings

Chevron packing in an open chamber

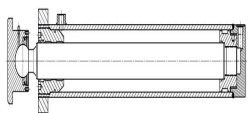


Chevron packing in a closed
chamber with support ring

Before mounting all of the single parts of the seal unit are to be greased. You can use mineral greases as long as these prove to be suitable for seals.

With rod diameters of 140 mm or more the chevron packings can be mounted in closed chambers. To compensate for the larger chamber a support ring is to be inserted.

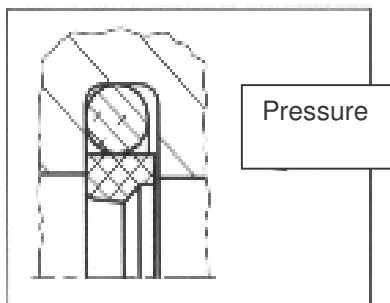
Support ring with closed slot



Assembly, Start-up and Maintenance of Hydraulic Cylinders

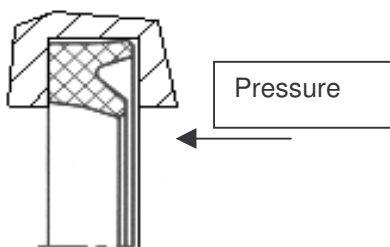
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6.1.2 Glydrings



When mounting the Glydring seals the O-ring is to be positioned in the mounting groove and then the profile is deformed to assume a kidney-like shape and snaps into place. Make sure that the profile ring has no sharp bends and that the sealing edge is placed correctly according to the direction of the pressure exerted.

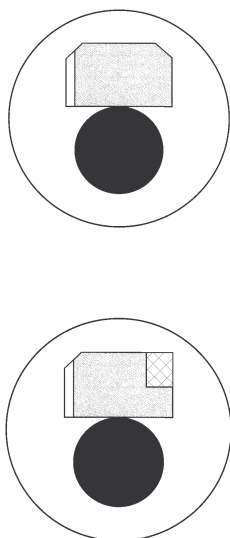
6.1.3 Grooved rings



The grooved rings can be mounted by hand in the recessed grooves. Make sure that the sealing lip is not damaged and that the sealing lip is placed correctly according to the direction of the pressure exerted.

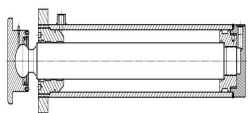
6.2 Piston Packings

6.2.1 Glydrings



Glydring piston packings can be mounted on solid pistons. The O-ring is mounted first and then the profile ring is pushed on. Warming the profile ring in hot water or with a jet of hot air at about 80 °C – 100 °C makes the mounting easier. The gauging of the seal can be performed with the relevant cylinder tube. This operation must be performed immediately after fitting.

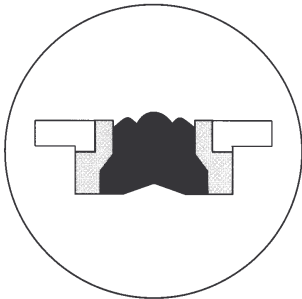
A special seal is the Glydring with gusset. The gusset is an additional, slotted support ring made of heat-resistant elastomer. When using this solution, follow the instructions provided in the assembly drawings.



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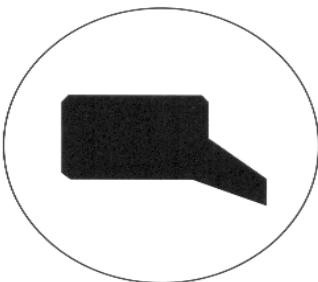
6.2.2 Compact Seals



Multi-unit piston packings can be snapped on for the most part with the aid of tools. First the rubber elastic contact element is to be snapped on. Then the slotted support rings and welded-angle rings are to be inserted.

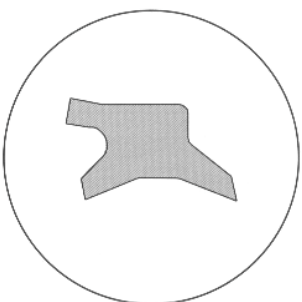
6.3 Scraper

6.3.1 Single-acting version

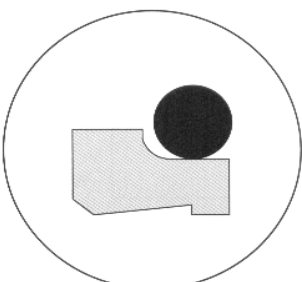


Generally speaking, the scrapers can be mounted quickly and easily thanks to their capacity to assume a kidney-like shape. The scrapers with metal retainers are pressed into an axially accessible housing space.

6.3.2 Double-acting version



One-part version



Two-part version

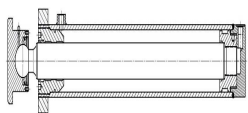
Double scrapers come into two different versions:

- One-part version (double-lip)
- Two-part version.

The thorough assembly of the equipment is the prerequisite for its proper functioning. Generally speaking, the scrapers can be mounted quickly and easily thanks to its capacity to assume a kidney-like shape.

The two-part double scrapers are composed of an oil control ring and an O-ring. Depending on the manufacturer even 2 O-rings can be used as the pretensioning element.

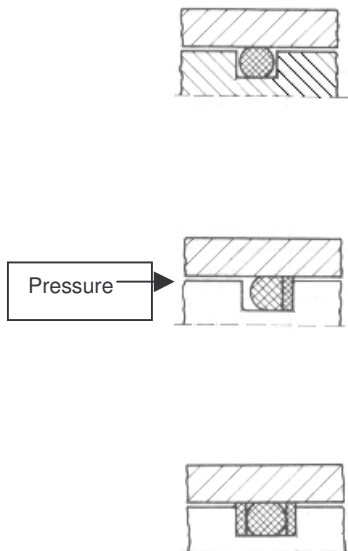
Follow carefully the assembly instructions in the drawings.



Assembly, Start-up and Maintenance of Hydraulic Cylinders

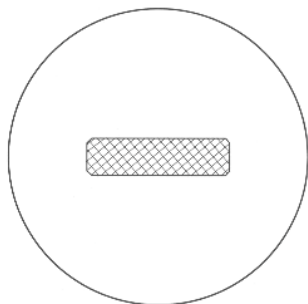
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6.4 O-rings



The O-rings must not be pulled over sharp edges or exposed shafts, threads, grooves, etc. It is recommended to cover the sharp spaces during assembly to prevent their damaging. The O-rings must not be twisted or drilled in the housing groove. They can be slightly stretched while being mounted. However, after mounting they must be left to rest to allow them to resume their original shape. To prevent extruding through the slot additional support rings can be mounted. Follow the instructions in the assembly drawing.

6.5 Guide Rings



Their purpose is that of guiding the piston and the rod, preventing metallic friction between the sliding components and supporting the resulting lateral forces. Follow the assembly instructions in the drawings.

7. Transportation and Storage

When handling the cylinder make sure that

- a) the cylinder is always completely run in to prevent the damaging of the piston rod
- b) Avoid knocking against the piston rod guide and the piston during handling to prevent the damaging of the guide elements and the piston rod.



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- c) The hydraulic connections are to be sealed with screw plugs or other suitable caps or covers with O-rings to prevent the infiltration of soiling

In case of storage for long periods of time all of the bare surfaces are to be coated with Tectyl.

In case of storage periods lasting more than 6 months the hydraulic cylinder is to be filled with a preservative oil and seal.

We recommend for instance RENOLIN MR 15 by Fuchs Lubritech GmbH.

When starting the cylinder there is no need to remove this oil from the cylinder chamber. This oil blends perfectly with all mineral oils as per DIN 51124.

The hydraulic cylinder is to be kept in a dry and dustless room free from corrosive substances and dampness.

If possible, the hydraulic cylinders are to be kept in a vertical position in case of long storage periods.

8. Safety Specifications

The safety requirements envisaged in standard EN 982, para. 5.3.2, in its most recent valid version are to be applied.